

RABBITMQ / .NET / IDEMPOTENCY

RabbitMQ redelivery in .NET

Dedupe before you process.

A consumer that rejects a redelivered message instead of reprocessing it.

OrderConsumer.cs / dedupe guard

CSHARP

```
public async Task HandleAsync(BasicDeliverEventArgs ea, CancellationToken ct)
{
    var messageId = ea.BasicProperties.MessageId;
    var key = $"processed:{messageId}";

    var isNew = await cache.StringSetAsync(
        key, "1", TimeSpan.FromHours(24), When.NotExists);

    if (!isNew)
    {
        channel.BasicAck(ea.DeliveryTag, false);
        return;
    }

    await ProcessOrderAsync(ea.Body, ct);
    channel.BasicAck(ea.DeliveryTag, false);
}
```

NOTE

Brokers guarantee at-least-once delivery, not exactly-once. Track processed message IDs with a short TTL so a redelivery is a no-op, not a duplicate order.

[EF CORE](#) / [POSTGRESQL](#) / [N+1](#)

N+1 hides until traffic finds it

Query once, not per row

Loading an order list should not fire one query per order for its line items.

BEFORE / lazy navigation per order

CSHARP



```
var orders = await db.Orders
    .Where(o => o.CustomerId == id)
    .ToListAsync(ct);

foreach (var o in orders)
{
    // fires one query per order
    var total = o.Items.Sum(i => i.Price);
}
```

AFTER / project what you need

CSHARP



```
var orders = await db.Orders
    .Where(o => o.CustomerId == id)
    .Select(o => new OrderSummary(
        o.Id, o.Items.Sum(i => i.Price)))
    .ToListAsync(ct);
```

NOTE

Lazy navigation looks correct in a unit test with three rows and falls over at production scale. Project or Include so EF Core issues one query, not N+1.

[NESTJS](#) / [.NET](#) / [VALIDATION](#)

Same contract, two runtimes.

Validate at the edge.

A CreateOrder DTO rejected before it reaches the domain layer, in Node and in .NET.

NestJS

TYPESCRIPT

```
class CreateOrderDto {  
  @IsUUID()  
  customerId: string;  
  
  @IsInt()  
  @Min(1)  
  quantity: number;  
}
```

.NET minimal API

CSHARP

```
public record CreateOrderDto(  
    [property: Required] Guid CustomerId,  
    [property: Range(1, int.MaxValue)] int Quantity  
);
```

A limit that trips itself.

limit_req needs burst.

Without burst and nodelay, a page load with six assets rate-limits itself.

No burst	NGINX 
<pre>limit_req_zone \$binary_remote_addr zone=api:10m rate=5r/s; location /api/ { limit_req zone=api; }</pre>	<pre>limit_req_zone \$binary_remote_addr zone=api:10m rate=5r/s; location /api/ { limit_req zone=api burst=12 nodelay; }</pre>
 Six parallel requests trip the limit for one user  503s show up in real traffic, not just load tests	 Absorbs a page's worth of concurrent calls  Still caps sustained abuse at 5r/s

A tight rate without burst rejects the same client's own concurrent asset requests, not just abusive traffic. Zone size and burst are what make the limit usable.

Healthy doesn't mean started

depends_on: service_healthy

Four services, four healthchecks — the API waits for real readiness, not just a running container.

postgresYAML

```
postgres:
  image: postgres:16
  healthcheck:
    test: ["CMD-SHELL", "pg_isready -U app"]
    interval: 5s
    retries: 5
```

redisYAML

```
redis:
  image: redis:7
  healthcheck:
    test: ["CMD", "redis-cli", "ping"]
    interval: 5s
    retries: 5
```

rabbitmqYAML

```
rabbitmq:
  image: rabbitmq:3-management
  healthcheck:
    test: ["CMD", "rabbitmq-diagnostics", "ping"]
    interval: 10s
    retries: 5
```

apiYAML

```
api:
  build: .
  depends_on:
    postgres:
      condition: service_healthy
    redis:
      condition: service_healthy
```

condition: service_started only waits for the container to launch. service_healthy waits for the healthcheck to pass, which is what dependent services actually need.

Same DI, less ceremony.

inject() in Angular

Constructor injection still works; the inject() function removes the boilerplate around it.

Constructor injection

TYPESCRIPT



```
export class OrderListComponent {
  constructor(
    private orders: OrderService,
    private auth: AuthService,
  ) {}
}
```

- ✗ Verbose for components with several dependencies
- ✗ Parameter order matters for readability

inject()

TYPESCRIPT



```
export class OrderListComponent {
  private orders = inject(OrderService);
  private auth = inject(AuthService);
}
```

- ✓ Reads top to bottom as plain field assignment
- ✓ Works in functional guards and resolvers too

inject() must run in an injection context — a constructor, a field initializer, or a factory function — not inside a later callback or setTimeout.

Name the role, not the colour.

Semantic design tokens

A stylesheet that says "accent" picks up the next palette release; one that says #2a5aa8 never will.

Raw values

CSS



```
.button {  
  background: #2a5aa8;  
  color: #ffffff;  
  border: 1px solid #87816f;  
}
```

Semantic tokens

CSS



```
.button {  
  background: var(--mk-color-accent);  
  color: var(--mk-color-on-accent);  
  border: 1px solid var(--mk-color-border-control);  
}
```

NOTE

Primitives hold raw values and never reach CSS; semantic tokens hold meaning and are all a site sees. Renaming one breaks consumers, changing its value does not.

CSS / DARK MODE

Follow the OS, allow an override Light and dark from one file

prefers-color-scheme picks the mode by default; a data-theme attribute on `<html>` forces one.

tokens.css

CSS

```
:root {
  --mk-color-bg: #F2F0EB;
  color-scheme: light;
}

@media (prefers-color-scheme: dark) {
  :root:not([data-theme="light"]) {
    --mk-color-bg: #0F141B;
    color-scheme: dark;
  }
}

:root[data-theme="dark"] {
  --mk-color-bg: #0F141B;
  color-scheme: dark;
}
```

- No attribute: the OS preference decides
- color-scheme switches native controls too

index.html

HTML

```
<html data-theme="dark">
  <head>
    <link
      rel="stylesheet"
      href="/v1/index.css"
    />
  </head>
  <body>
    ...
  </body>
</html>
```

- data-theme="light" or "dark" forces a mode
- Remove it to follow the OS again